



A Robust Segmentation Method Based on Improved U-Net

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Accepted: 16 May 2021 / Published online: 26 May 2021

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Abstract

Accurately reading spinal CT images is very important in clinical, but it usually costs some minutes and deeply depends on doctor's individual experiences. In this paper, we construct a scheme for spinal fracture lesions segmentation based on U-net, by introducing attention module, combining dilated convolution and U-net to get accurate lesions segmentation. First, we present four network schemes to compete in same data set, then get the best one, DU-net(dilated convolution), which replaces original convolution layer with dilated convolution in both contraction path and expansion path of U-net, to increase receptive field for more lesions feature information. Second, we introduce attention module to DU-net for accurate lesions segmentation by focusing on specific regions to improve lesions recognition of training model. Finally, we get prediction results by trained model of lesions segmentation on test data test. The experimental results show that our presented network has a better lesions segmentation performance than U-net, which can save time and reduce patients' suffering clinically.

Keywords U-net · Dilated convolution · Attention module · Segmentation

1 Introduction

With development of computer science technology, image segmentation by deep learning algorithms are widely applied in many applications, such as image recognition [1], image retrieval [2], medical image processing [3–6]. Yu, J., et al [1] devised a HDWE(Hierarchical Deep Word Embedding) model by integrating sparse constraints and an improved Relu oper-

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ator to address click feature prediction from visual features. Reference [2] proposed a novel ranking model based on the learning to rank framework, and utilize visual features and click features to obtain the ranking model. J. Yu, et al [3] proposed SPRNet(single-pixel reconstruction net), which performs efficient instance segmentation by introducing SPR(single-pixel reconstruction branch) to off-the-shelf one-stage detectors. Yu,Q.,et al [4] presented a cascaded trainable segmentation model termed as Crossbar-Net to accurately segment the kidney tumor in CT images. Wang,Z.,et al [5] proposed a novel approach to incorporate shape priors into a hierarchical learning based model, in order to segment organs at risk from head and neck CT images for the treatment of head and neck cancer. Wang,S.,et al [6] proposed a novel deep framework to segment male pelvic organs in CT images with incomplete annotation delineated in a very user-friendly manner. In this paper, we segment spinal fracture lesions by deep learning algorithm.

Spinal CT(Computed Tomography) images are acting pivotal role in treatment of spinal diseases, it is the main way for doctors to check, diagnose and locate spinal lesions. But there are some artificial errors and deviations in diagnosing, such as reliant on doctors' clinical experiences and subjective opinions, and it costs some time for viewing CT images. We propose a scheme to segment spinal lesions in CT images based on U-net [7], which can reduce time for orthopedic surgeons and diagnose objectively.

The present segmentation algorithms are either suitable for a concrete class vertebra, or have not a balance in detection correctness and detection speed, besides there are some manual work need to do. It will raise patients' suffering and increase doctors' diagnosis time. In this paper, we give a novel method based on U-net, which can keep focus on lesions region and augment receptive field of CT images, in order to collect more feature information of spinal lesions to provide assistance in diagnosis while keeping normal detection velocity. The main contributions in this paper are:

1. We present a model which combining dilated convolution and U-net to achieve more lesions feature information to prompt segmentation performance.
2. We present four novel network models, choose DU-net(dilated convolution) with best performance after experiments.
3. We introduce attention module to DU-net for keeping focus on lesions region to accurately segment spinal lesions.

Our paper is organized as follows: in Sect. 2, we describe the segmentation progress in deep learning . In Sect. 3, the materials used in this paper are described. In Sect. 4, we describe the improvements of proposed network mentioned above. Section 5 is our experiments and discussions. Finally, we present a conclusion of our work and next target in future.

2 Related Work

The purpose of spinal lesions segmentation is to identify and locate lesions in CT images, there are more and more intelligent algorithms for spinal lesions segmentation in recent years. We divide these algorithms into traditional machine learning methods and deep learning methods on basis of learning model, which are describe as follows:

2.1 Traditional Machine Learning Algorithms

Many researchers presented their contributions in diagnosing spinal diseases by machine learning methods. Mikulka, J.,et al [8] used a spine model for computer based simulation

of changes in the stress on the spine during idiopathic and syndromic deformity correction via vertebral osteotomy. Mandal, I. [9] adduced new hybrid machine learning ensembles for improving the performance of a computer aided diagnosis system integrated with multi method assessment process and statistical process control, used for the spine diagnosis based on noninvasive panoramic radiographs. Tej,M.S.,et al [10] utilized a previously proposed taxonomy that separated reproducibility into 3 components: technical, statistical, and conceptual reproducibility, and discussed common errors that lead to poor reproducibility, highlight the importance of generalizability when evaluating a ML model's performance, and provided suggestions to optimize generalizability to ensure adequate performance. Nam,K.H.,et al [11] presented a machine learning model in the spine research field, which could predict the T-score and osteoporotic vertebrae solely by measuring the HU of conventional CT, and this would help spine surgeons not to under-estimate the osteoporotic spine preoperatively. Natalia,F.,et al [12] presented the results of annotating lumbar spine MRI(Magnetic Resonance Imaging) images for automatic image segmentation and proposed confidence and consistency metrics to measure the quality and variability of the resulting ground truth data, respectively. Khan,O.,et al [13] developed machine learning algorithms predicting phenotypes of mild myelopathy patients that would benefit most from surgery, which showed good predictive power and provided information about the phenotypes of mild DCM(Degenerative Cervical Myelopathy) patients most likely to benefit from surgical intervention. Omar,K.,et al [14] successfully used ML(Machine Learning) to predict worse functional status after surgery for DCM and to determine associated predictors. Lee,S.,et al [15] evaluated machine learning models for identifying individuals with abnormal BMD(bone mineral density) through an analysis of spine X-ray features extracted by deep learning to alert high-risk osteoporosis populations. Shuang,Y.,et al [16] proposed a feature extraction and machine learning method for the classification of ultrasound images obtained from lumbar spine of pregnant patients in the transverse plane.

Although the algorithms mentioned above can process spinal lesions automatically by machine learning methods, the performance of robustness, segmentation precision and segmentation velocity are not very good, and the extracting networks for lesions feature information in these algorithms usually need careful devised.

2.2 Deep Learning Algorithms

The algorithms based on deep learning become popular in spinal lesions segmentation, for the performances of high segmentation accuracy, fast segmentation speed, and shared pre-trained model. Wang,Z.,et al [17] proposed a novel and efficient framework for medical image segmentation, which studied convolutional attention modules for feature refinement on all scale decoders and researched an ACNLL(adaptive clean noisy label learning) strategy based on the training process from under fitting to over fitting. H Tang,et al [18] developed an automatic lumbar spinal CT image segmentation method to assist LSS(lumbar spinal stenosis) diagnosis, which used a DDU-Net(dual densely connected U-shaped neural network) to segment spinal canal, dural sac and vertebral body in an end-to-end manner. Li,H.,et al [19] presented an lumbar spine magnetic resonance imaging image segmentation method based on deep learning, which contained multi-scale feature extraction based on three 3×3 convolution operators and vital information selection based on attention mechanism. Sewon,et al [20] proposed a new deep learning network capable of successfully segmenting intervertebral discs and their complex boundaries from MR(Magnetic Resonance) spine images. Huang,J.,et al [21] developed a deep learning based program (Spine Explorer) for automated segmentation

and quantification of the vertebrae and intervertebral discs on lumbar spine MRIs. Korez, R., et al [22] described a novel framework for automated spine and vertebrae detection and segmentation from 3-D CT images then used the obtained location of the whole spine to further detect the location of individual vertebrae within the spinal column. Lessmann, N. [23] presented an automatic method based on iterative convolutional neural networks which utilized the inherent order of the vertebral column to simplify the detection problem. Arif, Smmra [24] proposed a deep learning-based fully automatic framework for segmentation of cervical vertebrae in X-ray images towards building an automatic injury detection system. Ebrahimi, S. [25] aimed toward automated detection of specific vertebrae landmarks in spine radiographs, enabling automated adjustments. Chen, Y., et al [26] proposed a framework to capture the long-range contextual information in CT volumes in an end-to-end manner at the spine level. Wang, X [27] proposed an automatic method to localize and identify vertebrae by combining deep SSAE (stacked sparse auto encoder) contextual features and SRF (structured regression forest) to overcome the disadvantages of the approaches employing hand-crafted, low-level features. Li, S [28] proposed aco-attention module to learn the localization-guided segmentation attention and segmentation-guided localization attention based on combining a dilation convolution group and LSTM to learn the prior knowledge that the identification information was always in a fixed order for spine images. Shi, D., et al [29] developed a two-step algorithm to localize and segment just vertebral bodies by taking the advantage of the intensity pattern along the front spinal region, as well as GPU accelerations using convolutional neural networks. Lu, J. T., et al [30] developed an efficient methodology to leverage the subject-matter-expertise stored in large-scale archival reporting and image data for a deep-learning approach to fully-automated lumbar spinal stenosis grading. Rak, M., et al [31] proposed an automatic approach for fast vertebral body segmentation in three-dimensional magnetic resonance images of the whole spine.

Most of the algorithms mentioned above are based on U-net, which has achieved great progress in medical images process, but do not utilize enough features information of spinal fracture lesion CT images, In this paper, we give a way to combine dilated convolution and U-net to enhance receptive field of CT images for achieving thorough features in CT images.

3 Materials

We collect spinal CT images from XiJing Hospital (Military Medical University of Air Force) and other hospitals, randomly choose 30 cases of 3918 CT images for experiment. The data set is 7836 after data enhancement with pixels 256*256, and itemize as cfracture, tfracture, lfracture in the assistance of doctors. We splits data set with the ratio of 4:1, 70%, 5486 CT images are for training, 20%, 1568 CT images are for testing, and 10%, 784 for CT images are validation.

3.1 Data Preprocess

We preprocess the CT images to avoid errors and biases in CT images due to the CT images are checked by different CT equipments.

The CT images are DICOM (Digital Imaging and Communications in Medicine) format, which have a lot of information of spinal lesions. We make the CT images equalized due to there are some difference and bias in pixels caused by the different scanned CT equipments. Then we give a quantitative value HU (Hounsfield Unit) to process CT images, which reflects

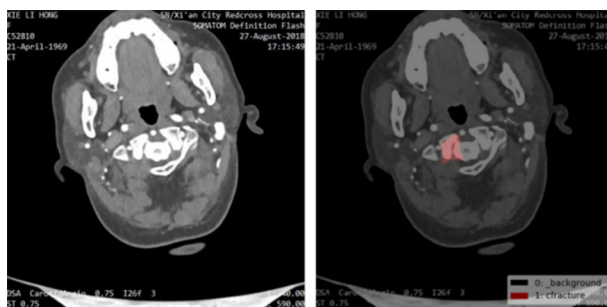


Fig. 1 cfraction lesions labelled. Zoomed-in for detail

the absorption of X-ray by organs and tissues. Each organ of human body has its own absorption coefficient. In CT image, black shadow represents low absorption area and low density area, such as lung, white shadow represents high absorption area and high density area, such as bone. Usually the bone HU value is larger than 400, so we take $[400, 2000]$ to normalize as $[0, 1]$. Then we eliminate noisy in CT images by Gauss Filter and get the ROI (Region of Interest) area through mathematical morphology to get lesions.

3.2 Data Augmentation

We randomly choose 30 series CT images which includes 3918 images for training and testing, because very large datasets are needed in deep learning, we expand data set by rotating image 90 degree clockwise, 180 degree clockwise, 270 degree clockwise, horizontal flip, vertical flip.

In data augmentation, we do not alter the appearances and locations of lesions, only turn lesions direction, but it is a different training sample in training, which is a very important and common way to expand data set in deep learning. After data augmentation, we can observe spinal lesions in different views, so as to get lesions information for correctly lesions segmentation.

3.3 Segment Spinal Lesions in CT Images

With the help of doctors in Xijing Hospital, we segment spinal lesions in each CT image using the software Labelme, label spinal lesions as cfraction (cervical fracture), tfraction (thoracic fracture), lfraction (lumbar fracture). Fig. 1, 2 and 3 is the sample segmentation results of cfraction, tfraction, lfraction.

4 Methodology

In this section, we describe improvements of proposed method based on U-net in detail.

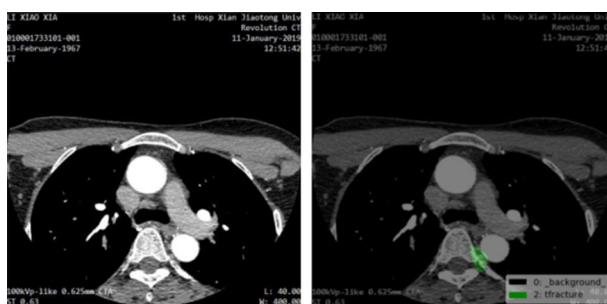


Fig. 2 tfracture lesions labelled. Zoomed-in for detail

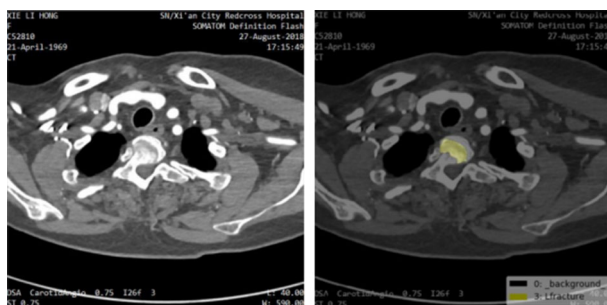


Fig. 3 lfraction lesions labelled. Zoomed-in for detail

4.1 U-net

Ronneberger et al. [7] gives the definition of U-net in 2015, which is novel scheme of convolution neural network and is appropriate for medical images segmentation. The U-net is based on FCN(Fully Convolutional Network)J. Long et al. [32] and is proved that it has more better medical image segmentation effect than FCN by experiments, which is shown in Fig. 4.

The U-net divide to contracting path(the encoder in left) and expanding path(the decoder in right).In left(the contracting path), there are four down-sampling blocks, which consists of two 3*3 convolution layers(without padding) and one max pooling layers (2*2 convolution, stride = 2), are used to extract the features of medical images and are used for the down-sampling operation. The feature channels of medical images are twice and the feature maps size are halved.

In right(the expanding path), there are also corresponding four up-sampling block, which includes two 3*3 convolution layers (without padding) and one de-convolution layer. The copy and crop which is shown in Fig. 4, connects the feature maps obtained by different layers for up-sampling block. Correspondingly, the feature channels of medical images are halved and the feature maps size are twice in up-sampling block. Then we map feature vector of 64 channels to different classified results in the last one 1*1 convolution layer.

We utilize the Relu(Rectified Linear Unit) function to concatenate the output feature maps of different shallow and deep layers, in order to get more accurate segmentation in medical images. Due to there is no FC(Full Connected) layer in U-net, and using valid convolution operation in training, which assure the input size and output size is not need to be same for very large medical images.

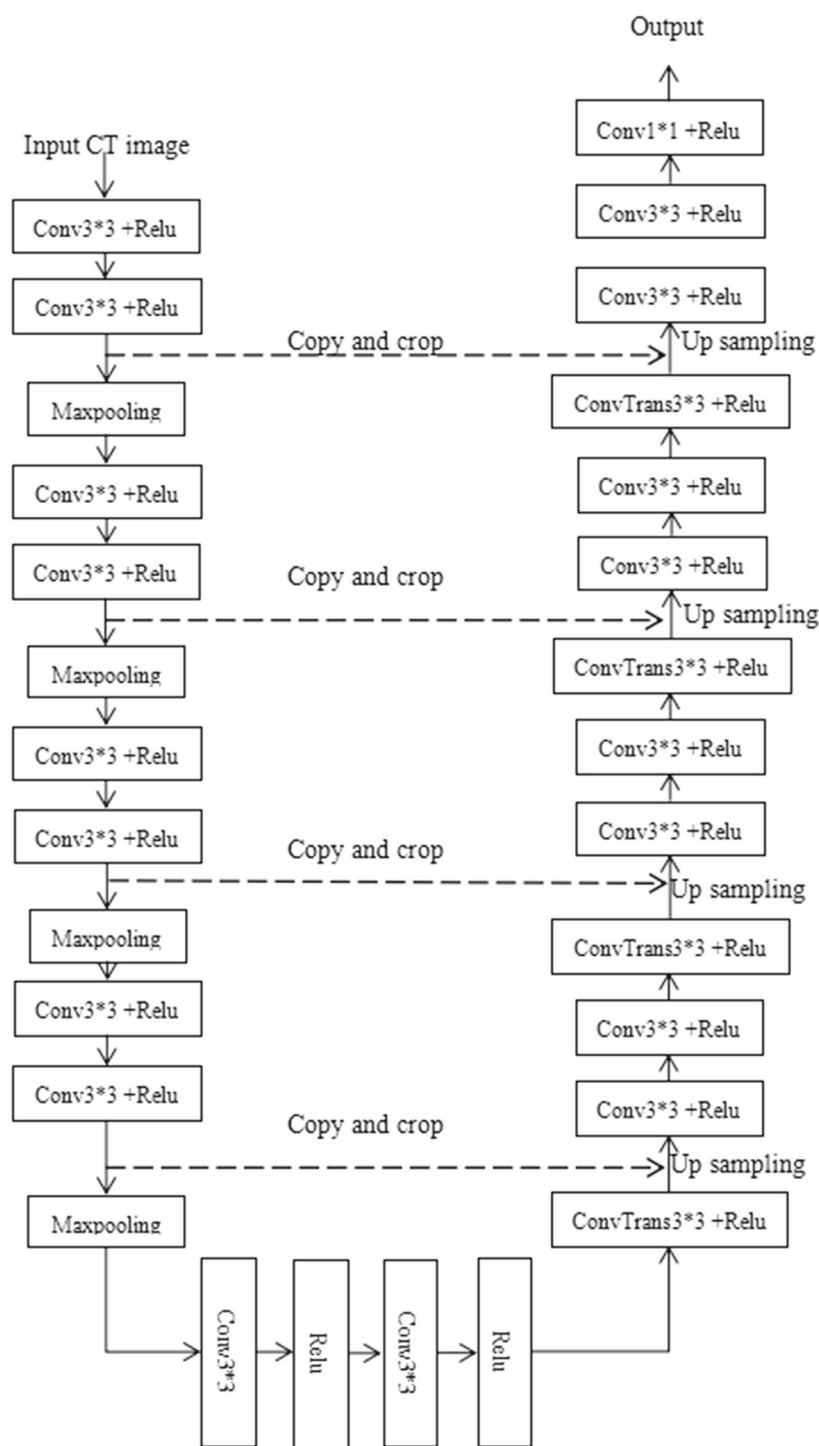


Fig. 4 The U-net architecture. Zoomed-in view for more details

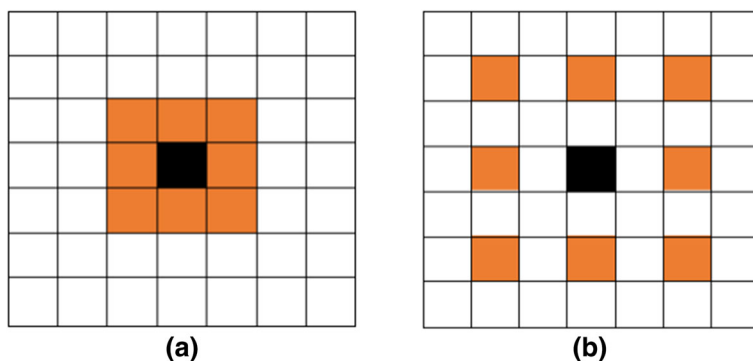


Fig. 5 Dilated convolution: (a) $r = 1$, conv kernel 3×3 (b) $r = 2$, conv kernel 5×5

4.2 Dilated Convolution

The dilated convolution is presented by Yu and Koltun [33], which is widely used in semantic segmentation and object detection. Dilated convolution can augment more information to the receptive field without introducing extra complexity to the network. Also, dilated convolution can exploit a thorough utilized information of original CT image to obtain more accurate lesion segmentation. The basic dilated convolution is illustrated in Fig. 5, here the convolution kernel size is 3×3 , and the dilated rate is 1. The kernel size of 1-dilated convolution is also 3×3 after zero pixel is inserted to CT image for a large receptive field; the kernel size of 2-dilated convolution is 5×5 , while the dilated rate is 2.

After utilizing dilated convolution in a traditional convolutional layer, we suppose the dilated rate is r , the kernel size of convolutional filter is k , so the receptive field R_k is:

$$R_k = (k - 1) \times (r - 1) + k \quad (1)$$

For example, if the kernel size of convolutional layer is 3×3 , the dilated rate r is 2, so the size of receptive field is 5.

Moreover, we will achieve a large receptive field by stacking multiple convolution layers, we suppose the two convolution layers H and $H-1$, and the filters are k_1 and k_2 , the receptive field size of $H+1$ layer R_{H+1} is:

$$R_{H+1} = k_1 + k_2 - 1 \quad (2)$$

For example, we have two stacked convolutional layers, with different filter size 3 and 5 respectively, then the receptive field size of stacked convolutional layers is 7. We will utilize three dilated convolutional layers with different dilated rates in our later experiments for a accurately spinal lesions segmentation.

4.3 Combine Dilated Convolution in U-net

In U-net, there is a problem of missing boundary of conventional unpadded every 3×3 convolution, to solve this problem, we propose four network models which are shown in Fig. 6, 7, 8 and 9.

In Fig. 6 we replace the conventional 3×3 convolution of the fourth down sampling path and corresponding up sampling instead of 2-dilate rate convolution, which is 1-PD-Unet(Portion

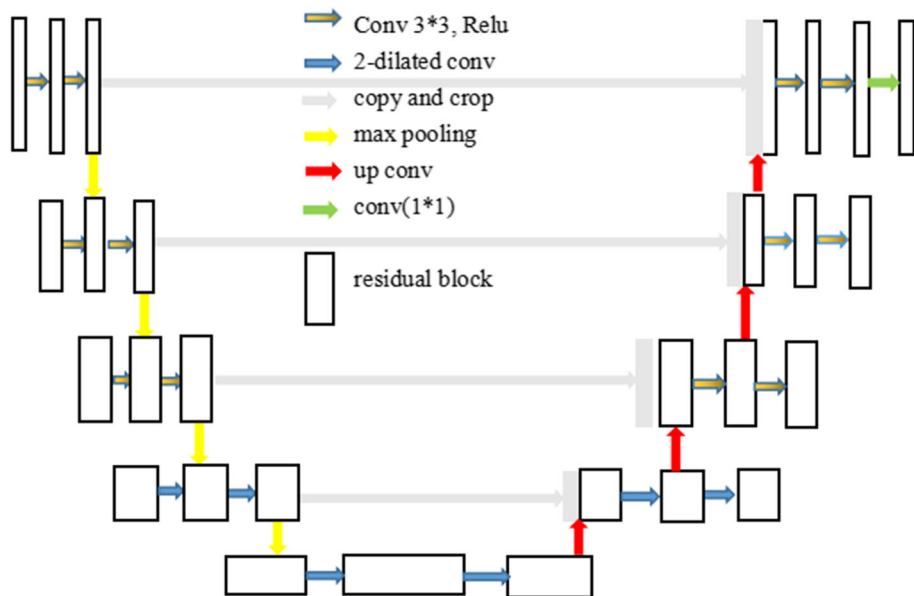


Fig. 6 1-PD-Unet. Zoomed-in view for more details

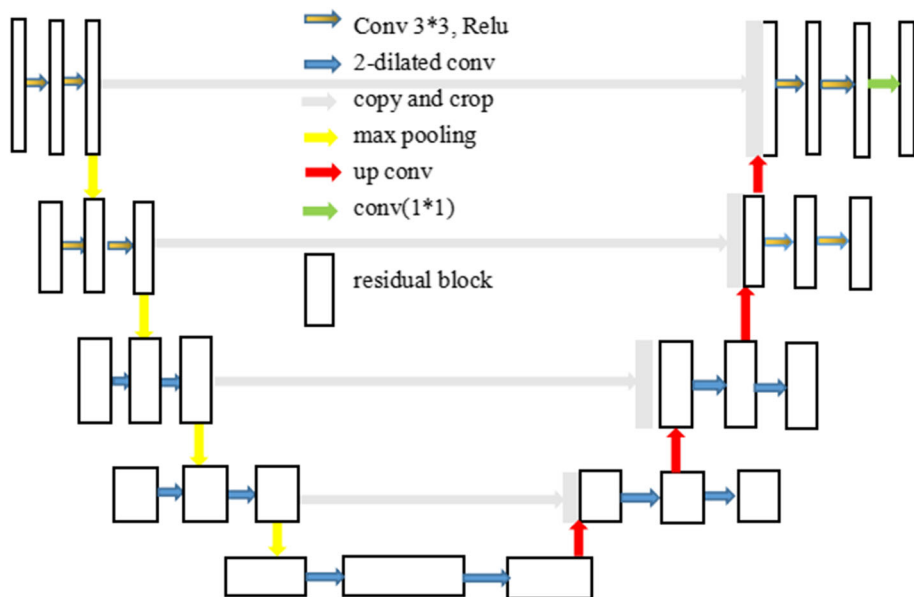


Fig. 7 2-PD-Unet. Zoomed-in view for more details

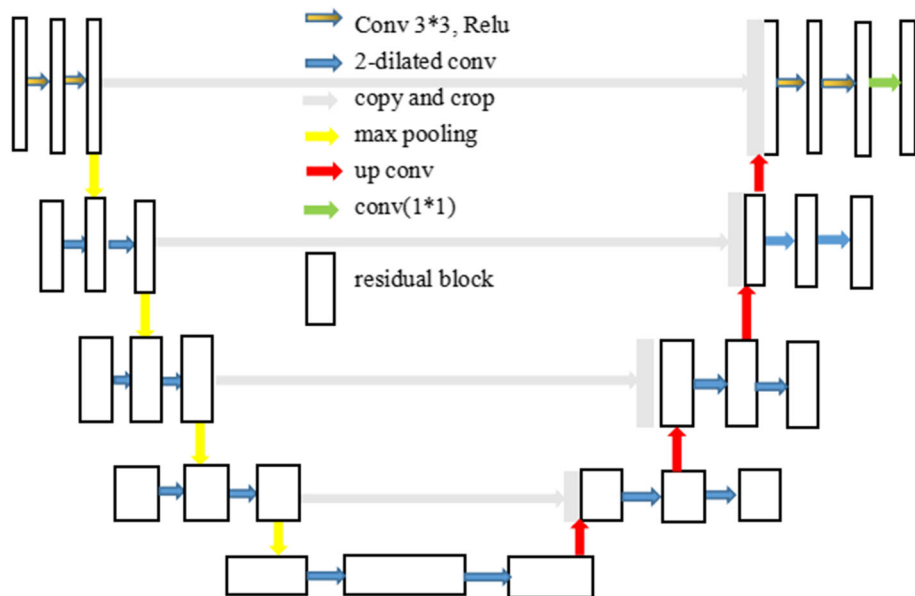


Fig. 8 3-PD-Unet. Zoomed-in view for more details

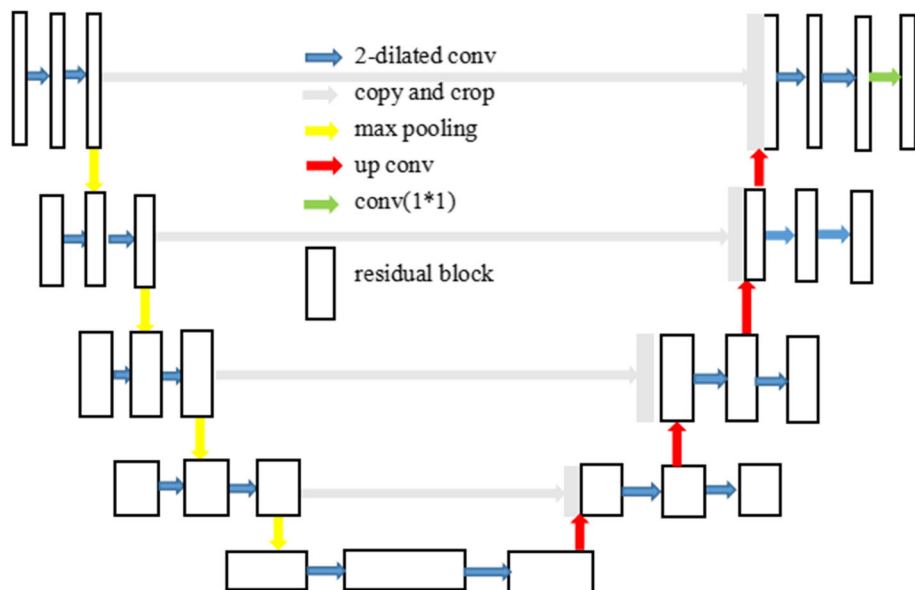


Fig. 9 Our proposed DU-net(Dilated convolution). Zoomed-in view for more details

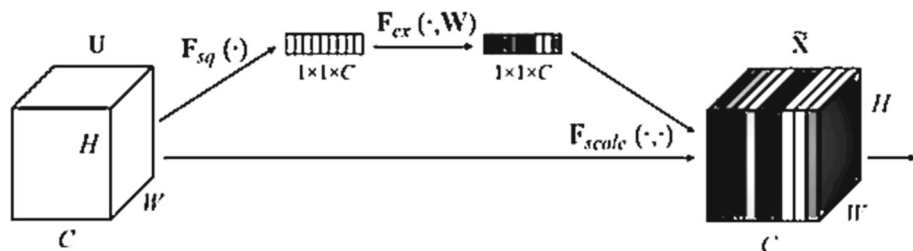


Fig. 10 The Structure of SE Module [34]

dilated convolution). In Fig. 7 we replace the conventional 3×3 convolution of both the third and fourth down sampling path and corresponding up sampling instead of 2-dilate rate convolution, which is 2-PD-Unet(Portion dilated convolution). In Fig. 8 we replace the conventional 3×3 convolution except the first down sampling path and corresponding up sampling instead of 2-dilate rate convolution, which is 3-PD-Unet(Portion dilated convolution). In Fig. 9 we replace the conventional 3×3 convolution of all the down sampling path and corresponding up sampling instead of 2-dilate rate convolution, except the last 1×1 convolution, which is DU-net(Dilated convolution). So we can get more CT images features for increase receptive field of CT images to segment lesions accurately.

4.4 Attention Module

Attention mechanism is widely used in various types of deep learning tasks, such as natural language processing, image recognition and speech recognition. It is one of the core technologies worthy of attention and deep understanding. For example, human beings can get the target area that we need to pay attention by scanning the global image quickly, i.e. the focus of attention, and then invest more attention resources in this area to obtain more detailed information of the target that we need, while suppressing other useless information. This is a way for human beings to use limited attention resources to quickly select high-value information from a large number of information. It is a survival mechanism formed in the long-term evolution of human beings. This attention mechanism greatly improves the efficiency and accuracy of visual information processing. The attention mechanism does not rely on output of previous layer, so we can process in parallel.

The spinal CT images include spinal lesions needed in this paper, as well as other regions of spine, soft tissue and background regions. So the useless information about spinal lesions should be filtered in training model, and focus on the worthy feature information of spinal lesions for accurate lesions segmentation. Therefore, we introduce attention mechanism to DU-net. We can divide attention mechanism into hard attention mechanism and soft attention mechanism. The hard attention mechanism pays attention in pixel level in image, each pixel is focused on, which is used in reinforcement learning. The soft attention mechanism pays attention to area or regions which are extracted in training network, and different weight of every area or region are calculated differentially.

The attention mechanism includes squeeze, excitation and attention. The SE(squeeze, excitation) module [34] is a kind of soft attention mechanism, which can evaluate the weight of every area or region while extracting feature information in training network, then it designate big weight to useful feature information and small weight to useless feature information, which is shown in Fig. 10.

In global average pooling, we suppose input x is (w, h, c) , then input x to SE module and obtain characteristic graph of size $C \times 1 \times 1$ of each region. The calculated operation is as follow:

$$z_c = F_{sq}(u_c) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W u_c(i, j). \quad (3)$$

Then there is operation of excitation, which includes two full connection layers and a Relu operation to measure dependence for each region, the operation is describe in Eq. 4:

$$s = F_{ex}(z, W) = \sigma(g(z, W)) = \sigma(W_2 \delta(W_1 z)). \quad (4)$$

Here W_1 is the parameter of the first FC, delta is Relu activation function, W_2 is the parameter of the second FC, and σ is sigmoid activation, then we get the parameter s of scale.

Finally, weight matrix s is designated to features information x . The different weights are corresponding to different regions, in order to make training model to focus on regions of useful and worthy feature information. so we can get in Eq. 5:

$$\tilde{x}_c = F_{scale}(u_c, s_c) = s_c u_c. \quad (5)$$

Therefore, the training can focus on regions of useful and worthy feature information while ignore the useless regions after adjusting the weights of different regions. Meanwhile, we can cut down the computation complexity of training network and improve accuracy of spinal lesions segmentation.

4.5 The Proposed Method

The whole proposed model is shown in Fig. 11, we introduce the attention modules and dilated convolution in both the compression path and expansion path. The proposed model inputs spinal CT images with size of 256×256 . We can extract more spinal lesions feature information by expanding receptive field in CT image after combining dilated convolution, and make training network focus on useful regions of spinal lesions to reduce the computation complexity and accurate lesions segmentation.

The network architecture of proposed model is shown in Table 1, from input to Conv5_2 are the encoder part. Then we introduce four attention layers in proposed model to get more favorable feature information for spinal lesions. In the decoder path, we also add four attention layers for accurately segment lesions.

5 Experiments and Discussions

In this section, we describe experimental results, comparative experiments to other algorithms. The configuration of experimental platform is shown in Table 2.

5.1 Evaluation Metrics

In this paper, we use formulas as follow to analyze the spinal fracture lesions segmentation of our experiments,

$$sensitivity = \frac{TP}{TP + FN}. \quad (6)$$

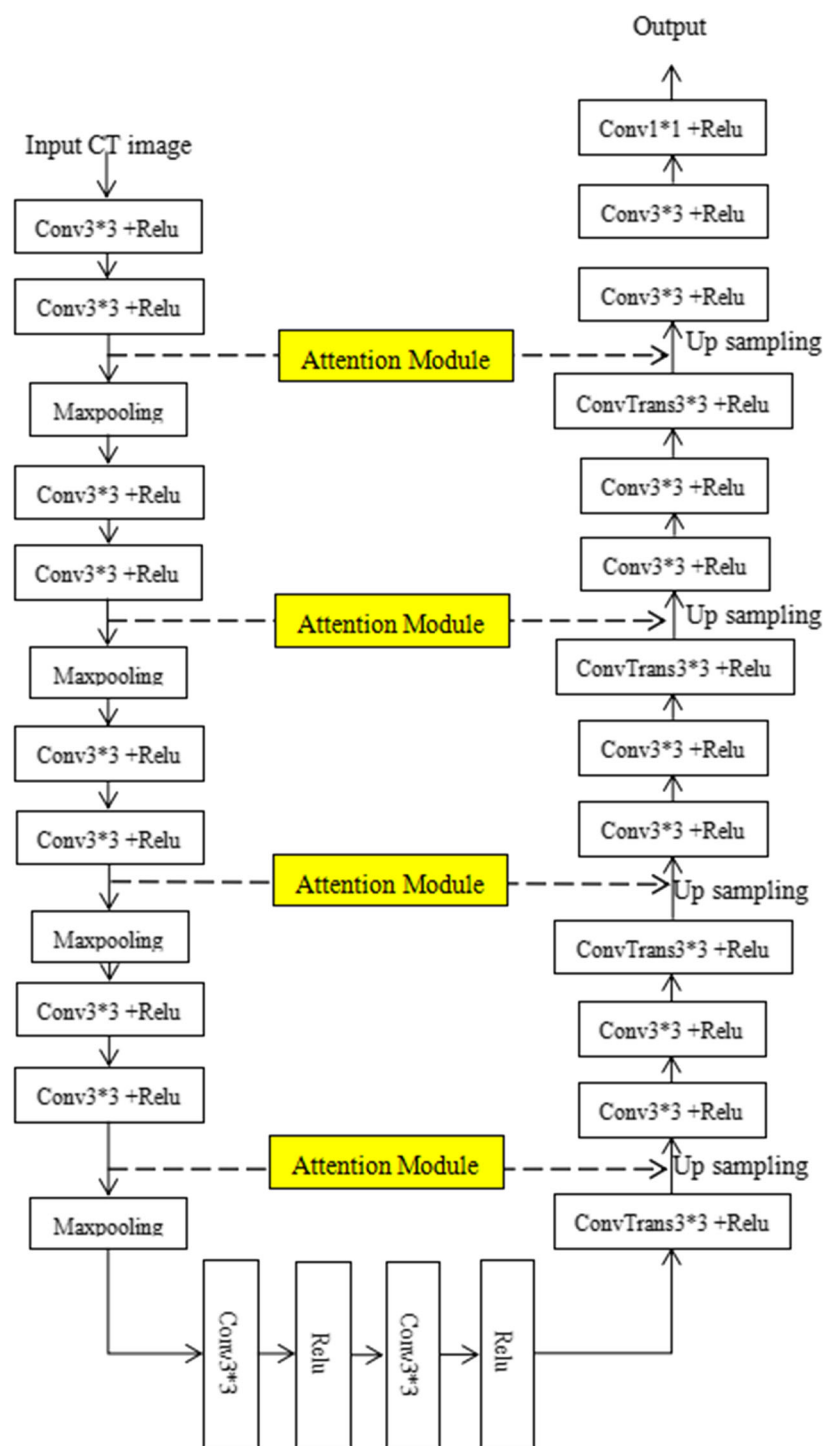


Fig. 11 The Structure of Proposed Model. Zoomed-in view for more details

Table 1 The Network Architecture of proposed model

Layer	Content	Ouput	Kenerl,Stride
Input	Input	256*256	
Conv1_1	Convolutional layer	256*256	3*3,1
Conv1_2	Convolutional layer	256*256	3*3,1
Down1	Max pooling	128*128	3*3,2
Conv2_1	Convolutional layer	128*128	3*3,1
Conv2_2	Convolutional layer	128*128	3*3,1
Down2	Max pooling	64*64	3*3,2
Conv3_1	Convolutional layer	64*64	3*3,1
Conv3_2	Convolutional layer	64*64	3*3,1
Down3	Max pooling	32*32	3*3,2
Conv4_1	Convolutional layer	32*32	3*3,1
Conv4_2	Convolutional layer	32*32	3*3,1
Down4	Max pooling	16*16	3*3,2
Conv5_1 (Bridge)	Convolutional layer	16*16	3*3,1
Conv5_2 (Bridge)	Convolutional layer	16*16	3*3,1
Up1	Transposed layer + Relu	32*32	3*3,2
Conv6_1	Convolutional layer	32*32	3*3,1
Conv6_2	Convolutional layer	32*32	3*3,1
Up2	Transposed layer + Relu	64*64	3*3,2
Conv7_1	Convolutional layer	64*64	3*3,1
Conv7_2	Convolutional layer	64*64	3*3,1
Up3	Transposed layer + Relu	128*128	3*3,2
Conv8_1	Convolutional layer	128*128	3*3,1
Conv8_2	Convolutional layer	128*128	3*3,1
Up4	Transposed layer + Relu	256*256	3*3,2
Conv9_1	Convolutional layer	256*256	3*3,1
Conv9_2	Convolutional layer	256*256	3*3,1
Attention1	Attention layer	256*256	3*3
Attention2	Attention layer	128*128	3*3
Attention3	Attention layer	64*64	3*3
Attention4	Attention layer	32*32	3*3
Output	Ouput	256*256	

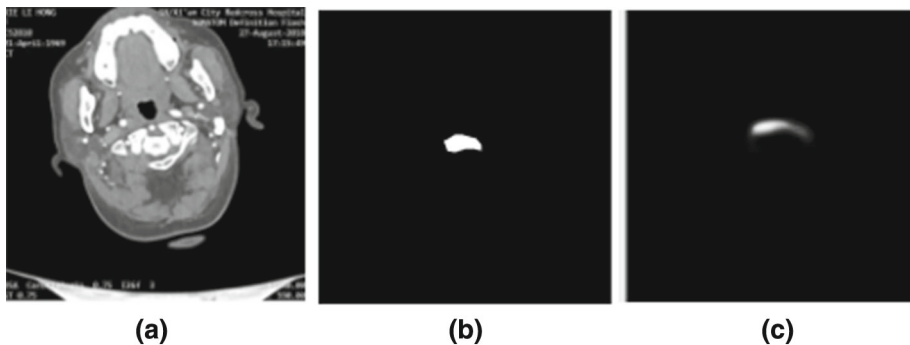
$$specificity = \frac{TN}{TN + FP}. \quad (7)$$

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \quad (8)$$

Here, TP is true positive which means the correctly detection lesions number. TN is true negative which means detection number of not spinal fracture lesion. FP is false positive which means the not spinal lesion number. FN is false negative which means the number of lesions detected not correctly. sensitivity means the percent of accurately spinal lesion prediction, itemize as spinal fracture lesion. specificity means the proportion of correctly

Table 2 The experimental platform

	Configuration
Operating system	Win 10
CPU	Intel Core i7-6700 @3.40GHz * 8
GPU	GTX1070, 8G
Hard disk	1T + 128G
Memory	32G
Framework	Tensorflow 1.8.0, keras 2.1.5
Language	Python 3.5
Library	Cuda 7.0, CUDNN, OpenCV, numpy 1.13.3
Learning-rate	1e-4

**Fig. 12** cfracture lesion segmentaion. **a** original CT image **b** ground truth **c** D-Unet

non-spinal lesion prediction results. accuracy means the correction rate of detected spinal lesion to the global CT image.

The (Dice similarity coefficient) f1 value is the average of the precision and recall which can represent the accuracy of spinal lesions segmentation, and its formula is defined as follows

$$f1 = \frac{2TP}{2TP + FP + FN}. \quad (9)$$

5.2 Experiments and Discussions

5.2.1 Experimental Results

Fig. 12 13 and 14 is the predicted segmentation results of spinal fracture(cfracture, tfraction, lfraction) lesion in proposed network,

We randomly select three groups results, including the predicted lesion of cervical fracture, thoracic fracture, lumbar fracture, (a) is original image, (b) is ground truth, (c) is predicted segmentation. The segmentation results are relatively accurate and effective, which can help doctor to diagnose spinal disease.

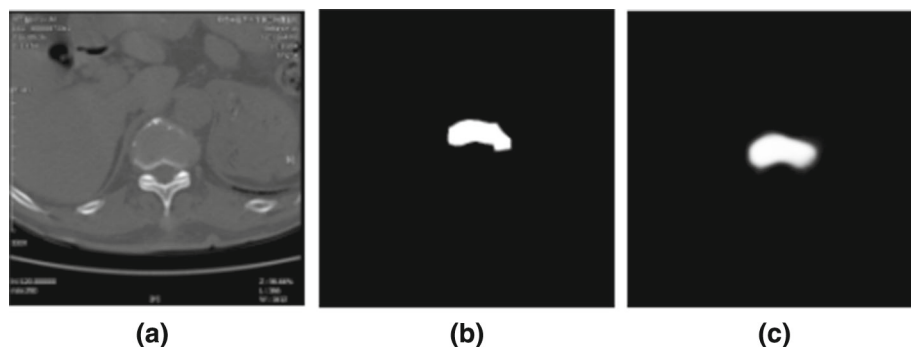


Fig. 13 tfracture lesion segmentaion. **a** original CT image **b** ground truth **c** D-Unet

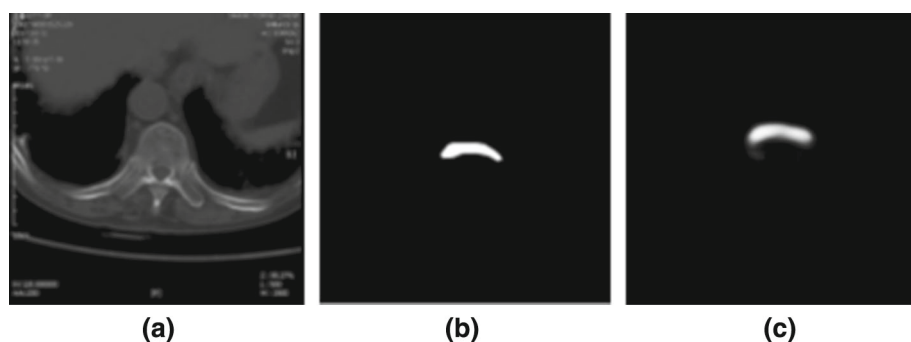


Fig. 14 lfracture lesion segmentaion. **a** original CT image **b** ground truth **c** D-Unet

Table 3 Lesion segmentation comparison of algorithms

Network	Sen	Spe	f1	Acc
U-net	0.818	0.952	0.915	0.965
1-PD-UNet	0.821	0.961	0.923	0.968
2-PD-UNet	0.823	0.963	0.935	0.970
3-PD-UNet	0.825	0.965	0.951	0.973
D-Unet	0.826	0.968	0.957	0.977
Proposed method	0.838	0.979	0.977	0.986

5.2.2 Comparison Between Dilated Convolutions

We get the best performance from the proposed four dilated U-net model in this experiment, we can get from Table 3, the sensitivity of U-net, 1-PD-UNet, 2-PD-UNet, 3-PD-UNet, D-Unet and proposed method is 0.818, 0.821, 0.823, 0.825, 0.826, 0.838 respectively, the specificity of U-net, 1-PD-UNet, 2-PD-UNet, 3-PD-UNet, D-Unet and proposed method is 0.952, 0.961, 0.963, 0.965, 0.968, 0.979 respectively, the accuracy of U-net, 1-PD-UNet, 2-PD-UNet, 3-PD-UNet, D-Unet and proposed method is 0.965, 0.968, 0.970, 0.973, 0.977, 0.986 respectively, the f1 value of U-net, 1-PD-UNet, 2-PD-UNet, 3-PD-UNet, D-Unet and proposed method is 0.915, 0.923, 0.935, 0.951, 0.957, 0.977 respectively. Therefore, the spinal lesion segmentation performance of proposed method is the best. The results shows the proposed method has a more accurately lesion segmentation.

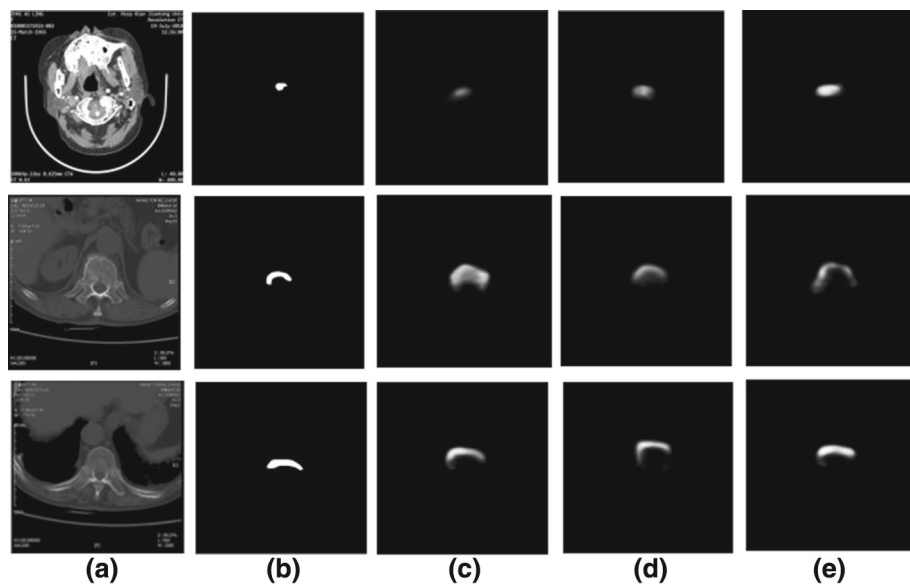


Fig. 15 The results, **a** original CT image **b** ground truth **c** Lu, Jentang, et al. [30] **d** U-net **e** proposed method. Zoomed-in view for more details

Table 4 Lesion segmentation comparison of algorithms

Network	Sen	Spe	f1	Acc	Time
Chuang et al. [35]	0.825	0.948	0.931	0.972	20 ms
Lu et al. [30]	0.822	0.951	0.927	0.970	19 ms
U-net	0.818	0.959	0.946	0.968	23 ms
D-Unet	0.826	0.968	0.957	0.977	22 ms
Proposed method	0.838	0.979	0.977	0.986	13 ms

The reason is that we make training model to pay more attention to useful regions of spinal lesions after introducing attention mechanism to DU-net, so we can get accurate lesions segmentation.

5.2.3 Comparisons to Other Algorithms

In this experiment, we give a comparison of our D-Unet and other algorithms in Fig. 15, (a) is original CT image, (b) is ground truth, (c) is Lu, Jentang, et al. [30], (d) is U-net, (e) is proposed method. We can see the segmentation performance of proposed method is more clearly and correctly than other algorithms, while Lu, Jentang, et al. [30] present only contour of spinal lesions and spinal lesions are segmented partly in traditional U-net, so the proposed method will give a solid foundation in diagnosing spinal diseases.

In Table 4, we present a quantitative comparison of proposed method and other algorithms, the sensitivity of Chuang et al. [35], Lu et al. [30], U-net, DU-net, and proposed method is 0.825, 0.822, 0.818, 0.826, 0.838 respectively, the specificity of Chuang et al. [35], Lu et al. [30], U-net, DU-net, and proposed method is 0.948, 0.951, 0.959, 0.968, 0.979 respectively, the accuracy of Chuang et al. [35], Lu et al. [30], U-net, DU-net, and proposed method is

0.972, 0.970, 0.968, 0.977, 0.986 respectively, the f1 value of Chuang et al. [35], Lu et al. [30], U-net, DU-net, and proposed method is 0.931, 0.927, 0.946, 0.957, 0.977 respectively, the elapsed time of Chuang et al. [35], Lu et al. [30], U-net, DU-net, and proposed method is 20 ms, 19 ms, 23 ms, 22 ms, 13 ms respectively. It can be observed that proposed method has a better performance in spinal lesion segmentation, and cost the least time.

The reason is that we not only increase the receptive field in CT images by combining dilated convolution, but also make training model to pay more attention to useful regions of spinal lesions after introducing attention mechanism to proposed method, so we can get accurate lesions segmentation.

6 Conclusion

Correctly segment spinal lesions is relatively significant in diagnosing spinal diseases. We present the proposed network which first combines dilated convolution and U-net to increase the receptive field of CT images, second introduce attention mechanism to proposed method for a better performance of spinal lesions segmentation. Through the experiments, we can see the proposed method has advantage in lesion segmentation than other algorithms, which will reduce the burden of doctors and the suffering of patients. Our next target is to segment spinal lesions by three-dimensional convolution neural network in future.

Acknowledgements The project in this paper is supported by Biomechanical Modeling of Lumbosacral Spine and Surgical Evaluation System”, Fund Number Nos. 61172147 and 61502365.

Author Contributions Junsheng Wu contributed to the conception of the study; Gang Sha performed the experiments, contributed significantly to analysis and manuscript preparation and the data analysis and wrote the manuscript; Bin Yu helped to perform the analysis with constructive discussions

Declarations

Conflict of interest The authors declare that they have no conflicts of interest.

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